

Scalable 3-Tier Web Architecture on AWS

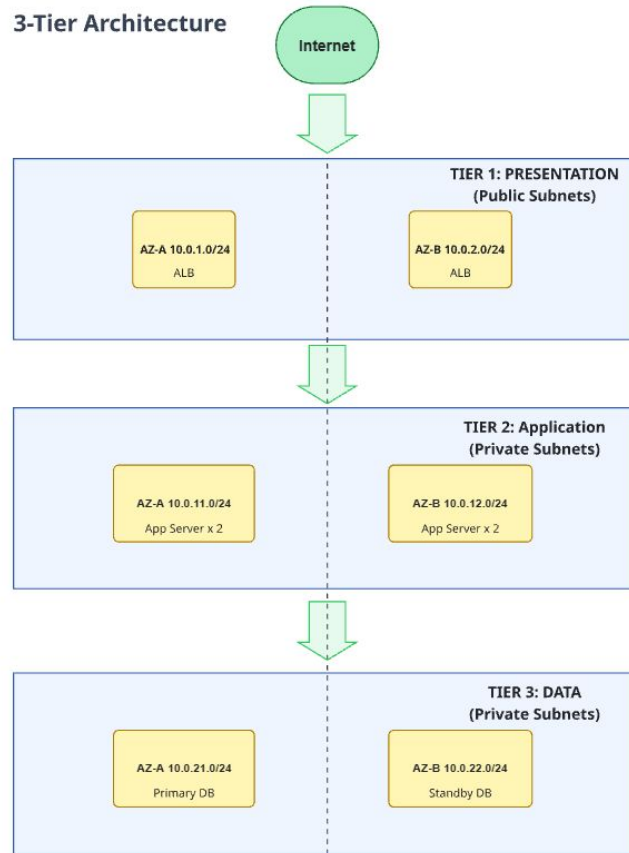
Highly Available Infrastructure with Automated CI/CD

Guangzheng Li

Architecture

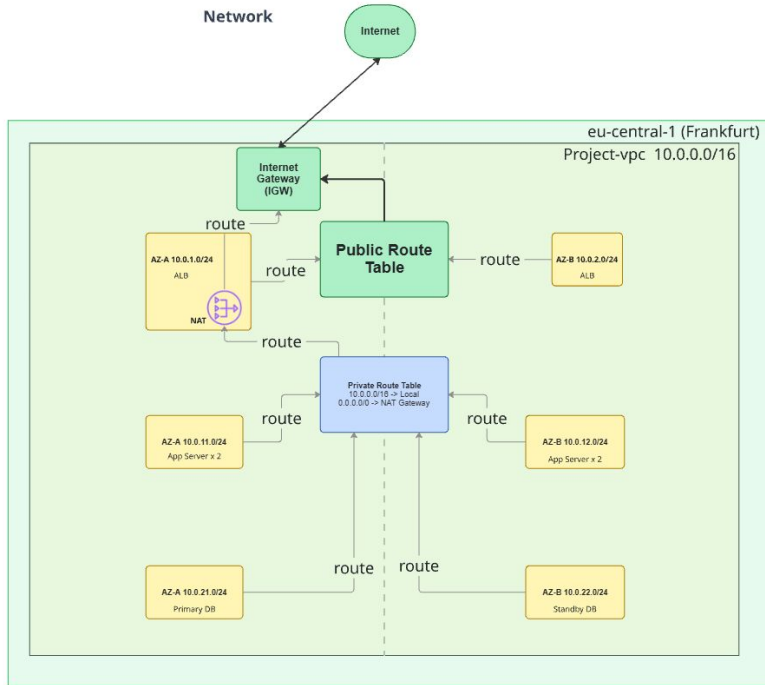
Architecture Diagram

3-Tier Architecture

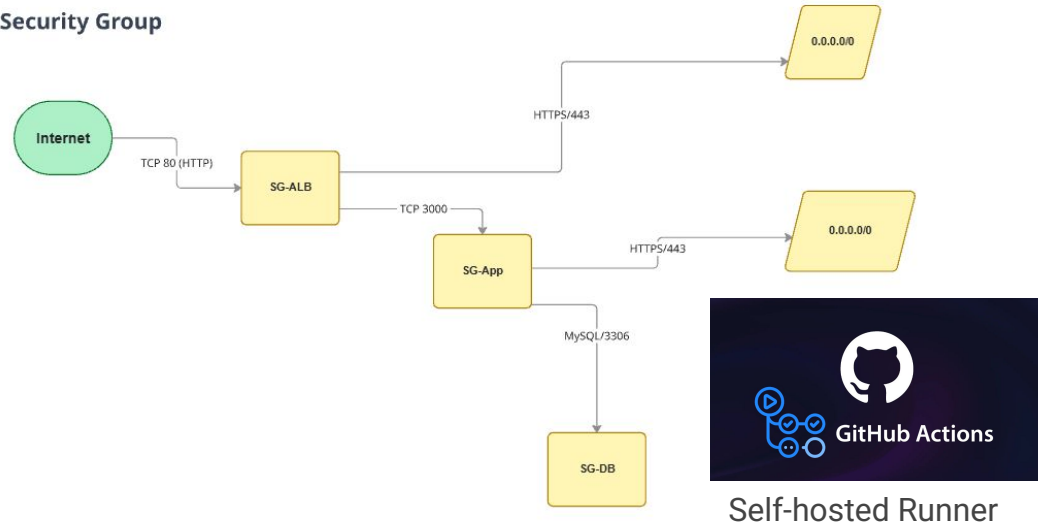


Architecture

Network & Security

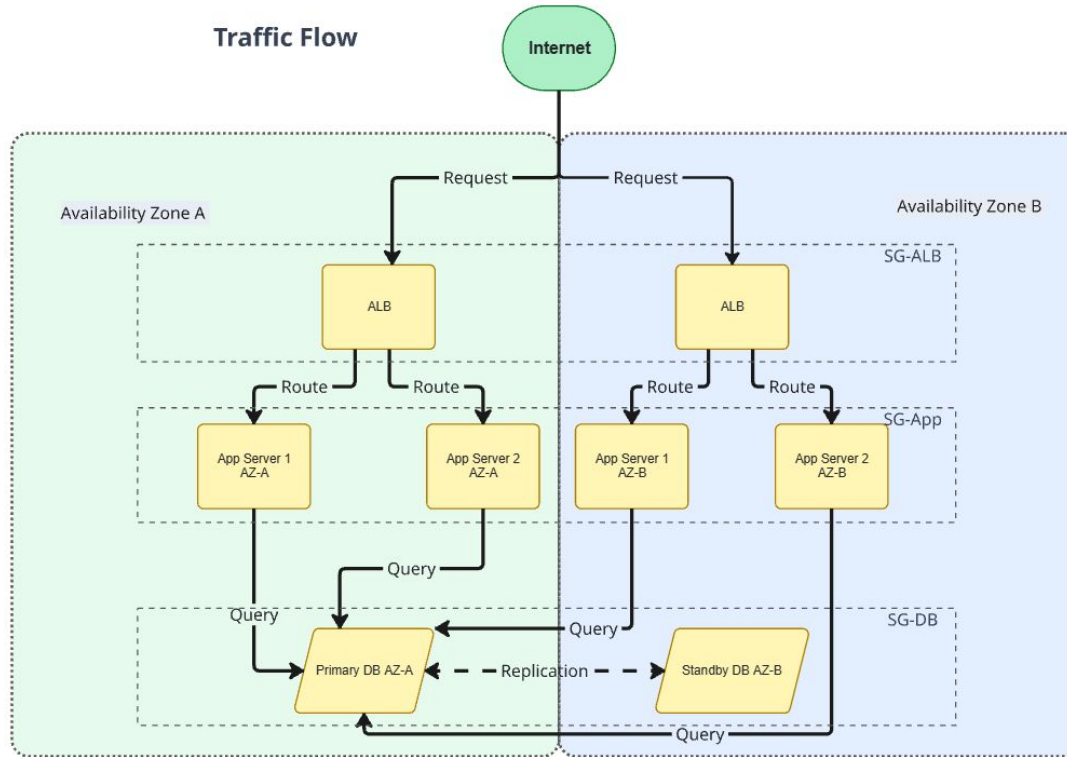


Security Group



Architecture

Traffic Flow



Demo

Project Demo

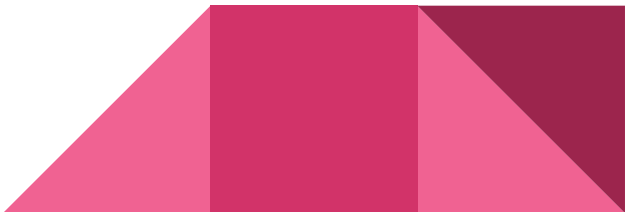
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Health Check

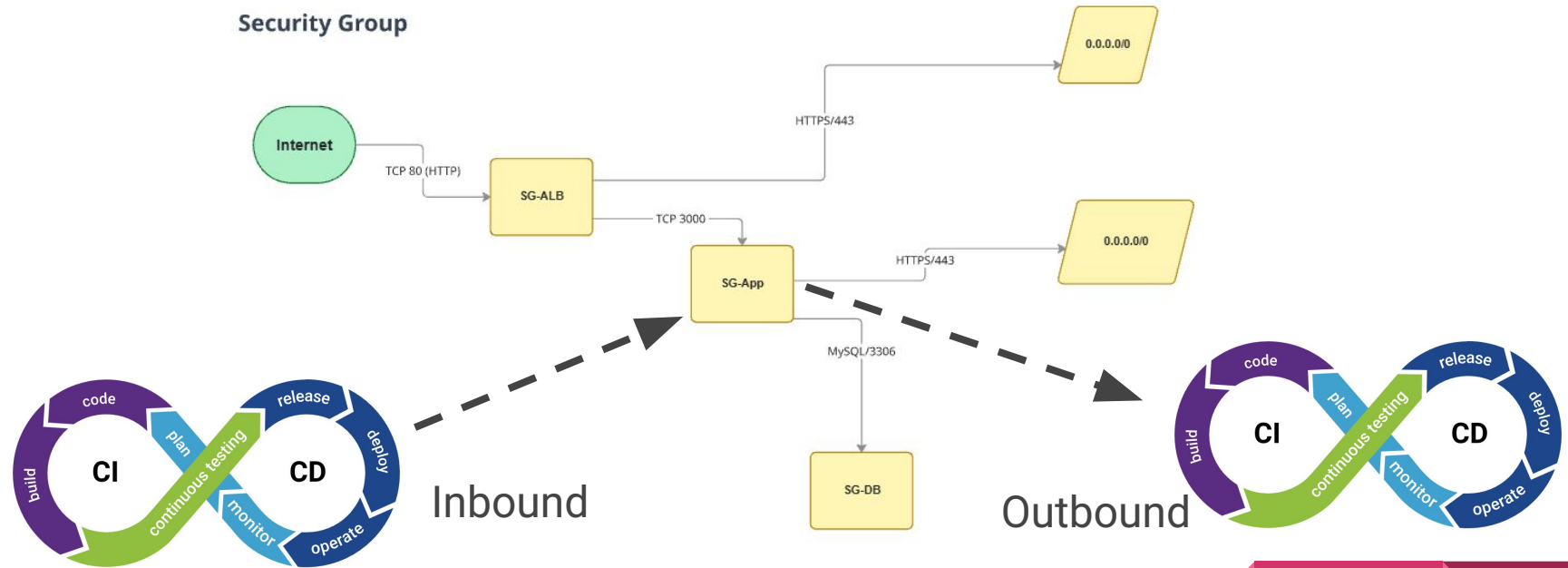
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User API

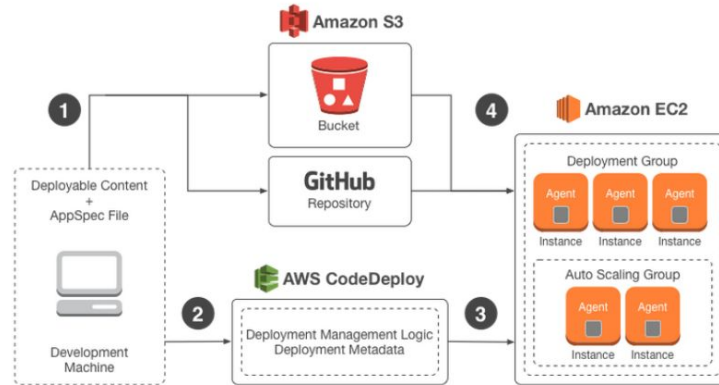
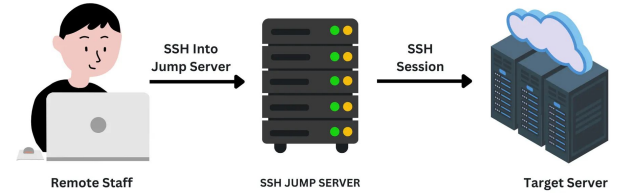
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Challenges & Solutions



Improvements & Next Steps



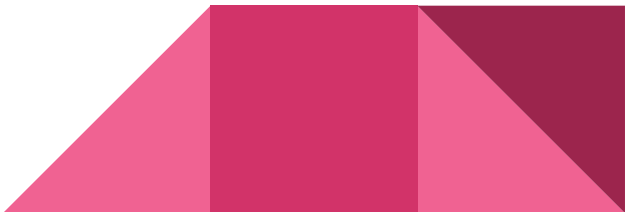
Cost Analysis

Free

- VPC & Subnets
- Security Groups & IAM
- Internet Gateway (IGW)
- EC2 (T2.micro/T3.micro)
- RDS (db.t2.micro/db.t3.micro)
- Amazon S3

Not Free

- NAT Gateway
- Application Load Balancer (ALB)
- IP (Public IPv4 Addresses)
- RDS Multi-AZ Deployment
- Data Transfer



Cost Analysis

1. NAT Gateway vs. VPC Endpoints

- **Current State:** Utilizing NAT Gateways to allow private instances to access GitHub and AWS APIs.
- **Optimization:** For a production-grade 3-tier app, we can replace NAT Gateways with **AWS VPC Endpoints (Interface Endpoints)**.
 - **Cost Impact:** Eliminates the hourly processing fee of NAT Gateways and the data transfer costs associated with the public internet.
 - **Security Bonus:** Traffic stays entirely within the AWS private backbone, further reducing the attack surface.

2. Dynamic Scaling & Right-Sizing

- **Current State:** Maintaining a fixed minimum number of instances in the Auto Scaling Group (ASG).
- **Optimization:** Implementing **Predictive Scaling** and **Scheduled Actions**.
 - **Cost Impact:** Automatically scales the fleet to zero or a minimum during non-business hours (e.g., nights and weekends).
 - **Right-Sizing:** Continuously monitoring CPU/RAM metrics to transition from **t3.micro** to even smaller or **Graviton-based instances (t4g.nano)** for significantly better price-performance ratios.

3. Multi-AZ Efficiency

- **Current State:** Ensuring high availability across multiple Availability Zones (AZs).
- **Optimization:** Minimizing **Cross-AZ Data Transfer** costs.
 - **Strategy:** Configuring the Load Balancer with **AZ-Affinity** (keeping traffic within the same AZ when possible) to avoid the \$0.01/GB cross-AZ surcharge.



Q&A

